```
#include <iostream>
using namespace std;
int binary_search(int a[],int n,int x)
{
   int first,last,mid;
   first=0;
   last=n-1;
   while(first<=last)
   {
      mid=(first+last)/2;
      if(a[mid]==x)
      return(mid);
      else if(a[mid]<x)
      first=mid+1;
     else last=mid-1;

   }
   return(-1);
}
int main()
{
   int n;
   int a[10],x;
   cout<<"how many element=";
   cin>>n;
   cout<<"enter "<<n <<" elements=";
   for( int i=0;i<n;i++)
   {
      cin>>a[i];
   }
   cout<<"Enter the element to be search=";
   cin>>x;
   int position=binary_search(a,n,x);
   if(position==-1)cout<<"Element not found";
   else cout<<"element found at position="<<position;
   return 0;
}

/*
PS D:\swarupa44> cd "d:\swarupa44\" ; if ($?) { g++ Binary_search.cpp -o Binary_search
} ; if ($?) { .\Binary_search }
how many element=9
enter 9 elements=10
20
```

```
30
40
50
60
70
80
90
Enter the element to be search=70
element found at position=6


PS D:\swarupa44> cd "d:\swarupa44\" ; if ($?) { g++ Binary_search.cpp -o Binary_search
} ; if ($?) { .\Binary_search }
how many element=5
enter 5 elements=45
35
25
15
55
Enter the element to be search=65
Element not found
*/
```